

**NIST IR 8477-Based Set Theory Relationship Mapping (STRM)**

**Reference Document:** Secure Controls Framework (SCF) version 2026.2

**STRM Guidance:** <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

**Focal Document:** [International Electrotechnical Commission \(IEC\) 62443 - 2 - 1:2024 - Security for industrial automation and control systems - Part 2 -](https://webstore.iec.ch/en/publication/62883)

**Focal Document URL:** <https://webstore.iec.ch/en/publication/62883>

**Published STRM URL:** <https://content.securecontrolsframework.com/strm/scf-strm-general-iec-62443-2-1-2024.pdf>

| FDE #      | FDE Name                                      | Focal Document Element (FDE) Description                                                                                                                                                | STRM Rationale | STRM Relationship | SCF Control                                           | SCF #    | Secure Controls Framework (SCF) Control Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Strength of Relationship | Notes |
|------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------|-------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------|
| ORG 1.1    | Information security management system (ISMS) | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Security, Compliance & Resilience Program (SCRp)      | GOV-01   | Mechanisms exist to facilitate the implementation of security, compliance and resilience governance controls.                                                                                                                                                                                                                                                                                                                                                                                                                                         | 10                       |       |
| ORG 1.2    | Background checks                             | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Equal             | Personnel Screening                                   | HRS-04   | Mechanisms exist to manage personnel security risk by screening individuals prior to authorizing access.                                                                                                                                                                                                                                                                                                                                                                                                                                              | 10                       |       |
| ORG 1.3    | Security roles and responsibilities           | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Defined Roles & Responsibilities                      | HRS-03   | Mechanisms exist to define cybersecurity roles & responsibilities for all personnel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 10                       |       |
| ORG 1.4    | Security awareness training                   | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Onboarding, Transferring & Offboarding Personnel      | HRS-01.1 | Mechanisms exist to proactively govern the following personnel management actions:<br>(1) Onboarding new personnel (e.g., new hires);<br>(2) Transferring personnel into new roles within the organization; and<br>(3) Offboarding personnel (e.g., termination of employment).                                                                                                                                                                                                                                                                       | 8                        |       |
| ORG 1.4    | Security awareness training                   | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Security, Compliance & Resilience Awareness Training  | SAT-02   | Mechanisms exist to provide all employees and contractors appropriate security, compliance and resilience awareness education and training that is relevant for their job function.                                                                                                                                                                                                                                                                                                                                                                   | 8                        |       |
| ORG 1.5    | Security responsibilities training            | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Role-Based Security, Compliance & Resilience Training | SAT-03   | Mechanisms exist to provide role-based security, compliance and resilience-related training:<br>(1) Before authorizing access to the system or performing assigned duties;<br>(2) When required by system changes; and<br>(3) Annually thereafter.                                                                                                                                                                                                                                                                                                    | 8                        |       |
| ORG 1.6    | Supply chain security                         | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Third-Party Management                                | TPM-01   | Mechanisms exist to facilitate the implementation of third-party management controls.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 10                       |       |
| ORG 2.1    | Security risk mitigation                      | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Supply Chain Risk Management (SCRM)                   | TPM-03   | Mechanisms exist to:<br>(1) Evaluate security risks and threats associated with Technology Assets, Applications and/or Services (TAAS) supply chains; and<br>(2) Take appropriate remediation actions to minimize the organization's exposure to those risks and threats, as necessary.                                                                                                                                                                                                                                                               | 8                        |       |
| ORG 2.1    | Security risk mitigation                      | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Risk Management Program                               | RSK-01   | Mechanisms exist to facilitate the implementation of strategic, operational and tactical risk management controls that are aligned                                                                                                                                                                                                                                                                                                                                                                                                                    | 10                       |       |
| ORG 2.1    | Security risk mitigation                      | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Risk Tolerance                                        | RSK-01.3 | Mechanisms exist to define organizational risk tolerance, the specified range of acceptable results.                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 8                        |       |
| ORG 2.1    | Security risk mitigation                      | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Risk Remediation                                      | RSK-06   | Mechanisms exist to remediate risks to an acceptable level.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 8                        |       |
| ORG 2.2    | Processes for discovery of security anomalies | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | No Relationship   | N/A                                                   | N/A      | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 0                        |       |
| ORG 2.2(a) | N/A                                           | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Automated Unauthorized Component Detection            | AST-02.2 | Automated mechanisms exist to detect and alert upon the detection of unauthorized hardware, software and firmware components.                                                                                                                                                                                                                                                                                                                                                                                                                         | 8                        |       |
| ORG 2.2(b) | N/A                                           | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Inbound & Outbound Communications Traffic             | MON-01.3 | Mechanisms exist to continuously monitor inbound and outbound communications traffic for unusual or unauthorized activities or conditions.                                                                                                                                                                                                                                                                                                                                                                                                            | 8                        |       |
| ORG 2.2(b) | N/A                                           | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Unauthorized Network Services                         | MON-11.2 | Automated mechanisms exist to detect unauthorized network services and alert incident response personnel.                                                                                                                                                                                                                                                                                                                                                                                                                                             | 8                        |       |
| ORG 2.2(c) | N/A                                           | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Vulnerability Scanning                                | VPM-06   | Mechanisms exist to detect vulnerabilities and configuration errors by routine vulnerability scanning of systems and applications.                                                                                                                                                                                                                                                                                                                                                                                                                    | 8                        |       |
| ORG 2.2(d) | N/A                                           | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Anomalous Behavior                                    | MON-16   | Mechanisms exist to utilize User & Entity Behavior Analytics (UEBA) and/or User Activity Monitoring (UAM) solutions to detect and respond to anomalous behavior that could indicate account compromise or other malicious activities.                                                                                                                                                                                                                                                                                                                 | 8                        |       |
| ORG 2.3    | Secure development and support                | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Technology Development & Acquisition                  | TDA-01   | Mechanisms exist to facilitate the implementation of tailored development and acquisition strategies, contract tools and procurement methods to meet unique business needs.                                                                                                                                                                                                                                                                                                                                                                           | 10                       |       |
| ORG 2.3    | Secure development and support                | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Secure Software Development Practices (SSDP)          | TDA-06   | Mechanisms exist to develop applications based on Secure Software Development Practices (SSDP).                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 8                        |       |
| ORG 2.4    | SP reviews                                    | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Information Assurance (IA) Operations                 | IAO-01   | Mechanisms exist to facilitate the implementation of security, compliance and resilience assessment and authorization controls.                                                                                                                                                                                                                                                                                                                                                                                                                       | 10                       |       |
| ORG 2.4(a) | N/A                                           | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Assessments                                           | IAO-02   | Mechanisms exist to formally assess the security, compliance and resilience controls in Technology Assets, Applications and/or Services (TAAS) through Information Assurance Program (IAP) activities to determine the extent to which the controls are implemented correctly, operating as intended and producing the desired outcome with respect to meeting expected requirements.                                                                                                                                                                 | 8                        |       |
| ORG 2.4(b) | N/A                                           | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Technical Verification                                | IAO-06   | Mechanisms exist to perform Information Assurance Program (IAP) activities to evaluate the design, implementation and effectiveness of technical security, compliance and resilience controls.                                                                                                                                                                                                                                                                                                                                                        | 8                        |       |
| ORG 2.4(c) | N/A                                           | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Threat Analysis & Flaw Remediation During Development | IAO-04   | Mechanisms exist to require system developers and integrators to create and execute a Security Testing and Evaluation (ST&E) plan, or similar process, to identify and remediate flaws during development.                                                                                                                                                                                                                                                                                                                                            | 8                        |       |
| ORG 2.4(d) | N/A                                           | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Security Assessment Report (SAR)                      | IAO-02.4 | Mechanisms exist to produce a Security Assessment Report (SAR) at the conclusion of a security assessment to certify the results of the assessment and assist with any remediation actions.                                                                                                                                                                                                                                                                                                                                                           | 8                        |       |
| ORG 3.1    | Physical access control                       | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Physical & Environmental Protections                  | PES-01   | Mechanisms exist to facilitate the operation of physical and environmental protection controls.                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 10                       |       |
| CM 1.1     | Asset inventory baseline                      | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Asset Inventories                                     | AST-02   | Mechanisms exist to perform inventories of Technology Assets, Applications, Services and/or Data (TAASD) that:<br>(1) Accurately reflects the current TAASD in use;<br>(2) Identifies authorized software products, including business justification details;<br>(3) Is at the level of granularity deemed necessary for tracking and reporting;<br>(4) Includes organization-defined information deemed necessary to achieve effective property accountability; and<br>(5) Is available for review and audit by designated organizational personnel. | 10                       |       |
| CM 1.1(a)  | N/A                                           | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Asset Inventories                                     | AST-02   | Mechanisms exist to perform inventories of Technology Assets, Applications, Services and/or Data (TAASD) that:<br>(1) Accurately reflects the current TAASD in use;<br>(2) Identifies authorized software products, including business justification details;<br>(3) Is at the level of granularity deemed necessary for tracking and reporting;<br>(4) Includes organization-defined information deemed necessary to achieve effective property accountability; and<br>(5) Is available for review and audit by designated organizational personnel. | 10                       |       |
| CM 1.1(b)  | N/A                                           | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Asset Inventories                                     | AST-02   | Mechanisms exist to perform inventories of Technology Assets, Applications, Services and/or Data (TAASD) that:<br>(1) Accurately reflects the current TAASD in use;<br>(2) Identifies authorized software products, including business justification details;<br>(3) Is at the level of granularity deemed necessary for tracking and reporting;<br>(4) Includes organization-defined information deemed necessary to achieve effective property accountability; and<br>(5) Is available for review and audit by designated organizational personnel. | 10                       |       |

| FDE #     | FDE Name                                                 | Focal Document Element (FDE) Description                                                                                                                                                | STRM Rationale | STRM Relationship | SCF Control                                                  | SCF #    | Secure Controls Framework (SCF) Control Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Strength of Relationship | Notes |
|-----------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------|--------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------|
| CM 1.1(c) | N/A                                                      | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Asset Inventories                                            | AST-02   | Mechanisms exist to perform inventories of Technology Assets, Applications, Services and/or Data (TAASD) that:<br>(1) Accurately reflects the current TAASD in use;<br>(2) Identifies authorized software products, including business justification details;<br>(3) Is at the level of granularity deemed necessary for tracking and reporting;<br>(4) Includes organization-defined information deemed necessary to achieve effective property accountability; and<br>(5) Is available for review and audit by designated organizational personnel. | 10                       |       |
| CM 1.1(d) | N/A                                                      | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Asset Inventories                                            | AST-02   | Mechanisms exist to perform inventories of Technology Assets, Applications, Services and/or Data (TAASD) that:<br>(1) Accurately reflects the current TAASD in use;<br>(2) Identifies authorized software products, including business justification details;<br>(3) Is at the level of granularity deemed necessary for tracking and reporting;<br>(4) Includes organization-defined information deemed necessary to achieve effective property accountability; and<br>(5) Is available for review and audit by designated organizational personnel. | 10                       |       |
| CM 1.1(e) | N/A                                                      | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Asset Inventories                                            | AST-02   | Mechanisms exist to perform inventories of Technology Assets, Applications, Services and/or Data (TAASD) that:<br>(1) Accurately reflects the current TAASD in use;<br>(2) Identifies authorized software products, including business justification details;<br>(3) Is at the level of granularity deemed necessary for tracking and reporting;<br>(4) Includes organization-defined information deemed necessary to achieve effective property accountability; and<br>(5) Is available for review and audit by designated organizational personnel. | 10                       |       |
| CM 1.1(f) | N/A                                                      | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Asset Inventories                                            | AST-02   | Mechanisms exist to perform inventories of Technology Assets, Applications, Services and/or Data (TAASD) that:<br>(1) Accurately reflects the current TAASD in use;<br>(2) Identifies authorized software products, including business justification details;<br>(3) Is at the level of granularity deemed necessary for tracking and reporting;<br>(4) Includes organization-defined information deemed necessary to achieve effective property accountability; and<br>(5) Is available for review and audit by designated organizational personnel. | 10                       |       |
| CM 1.1(g) | N/A                                                      | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Asset Inventories                                            | AST-02   | Mechanisms exist to perform inventories of Technology Assets, Applications, Services and/or Data (TAASD) that:<br>(1) Accurately reflects the current TAASD in use;<br>(2) Identifies authorized software products, including business justification details;<br>(3) Is at the level of granularity deemed necessary for tracking and reporting;<br>(4) Includes organization-defined information deemed necessary to achieve effective property accountability; and<br>(5) Is available for review and audit by designated organizational personnel. | 10                       |       |
| CM 1.1(h) | N/A                                                      | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Asset Inventories                                            | AST-02   | Mechanisms exist to perform inventories of Technology Assets, Applications, Services and/or Data (TAASD) that:<br>(1) Accurately reflects the current TAASD in use;<br>(2) Identifies authorized software products, including business justification details;<br>(3) Is at the level of granularity deemed necessary for tracking and reporting;<br>(4) Includes organization-defined information deemed necessary to achieve effective property accountability; and<br>(5) Is available for review and audit by designated organizational personnel. | 10                       |       |
| CM 1.1(i) | N/A                                                      | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Asset Inventories                                            | AST-02   | Mechanisms exist to perform inventories of Technology Assets, Applications, Services and/or Data (TAASD) that:<br>(1) Accurately reflects the current TAASD in use;<br>(2) Identifies authorized software products, including business justification details;<br>(3) Is at the level of granularity deemed necessary for tracking and reporting;<br>(4) Includes organization-defined information deemed necessary to achieve effective property accountability; and<br>(5) Is available for review and audit by designated organizational personnel. | 10                       |       |
| CM 1.2    | Infrastructure drawings/documentation                    | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Network Diagrams & Data Flow Diagrams (DFDs)                 | AST-04   | Mechanisms exist to maintain network architecture diagrams that:<br>(1) Contain sufficient detail to assess the security of the network's architecture;<br>(2) Reflect the current architecture of the network environment; and<br>(3) Document all sensitive and/or regulated data flows.                                                                                                                                                                                                                                                            | 10                       |       |
| CM 1.3    | Configuration settings                                   | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Secure Baseline Configurations                               | CFG-02   | Mechanisms exist to develop, document and maintain secure baseline configurations for Technology Assets, Applications and/or Services (TAAS) that are consistent with industry-accepted system hardening standards.                                                                                                                                                                                                                                                                                                                                   | 8                        |       |
| CM 1.3    | Configuration settings                                   | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Automated Central Management & Verification                  | CFG-02.2 | Automated mechanisms exist to govern and report on baseline configurations of Technology Assets, Applications and/or Services (TAAS) through Continuous Diagnostics and Mitigation (CDM), or similar technologies.                                                                                                                                                                                                                                                                                                                                    | 8                        |       |
| CM 1.4    | Change control                                           | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Change Management Program                                    | CHG-01   | Mechanisms exist to facilitate the implementation of a change management program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 10                       |       |
| CM 1.4    | Change control                                           | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Configuration Change Control                                 | CHG-02   | Mechanisms exist to govern the technical configuration change control processes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 8                        |       |
| NET 1.1   | Segmentation from non-IACS zones                         | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Network Segmentation (macrosegmentation)                     | NET-06   | Mechanisms exist to implement network segmentation within network architectures to isolate Technology Assets, Applications and/or Services (TAAS) from other network resources.                                                                                                                                                                                                                                                                                                                                                                       | 10                       |       |
| NET 1.2   | Documentation of zones and network zone interconnections | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Network Security Controls (NSC)                              | NET-01   | Mechanisms exist to develop, govern & update procedures to facilitate the implementation of Network Security Controls (NSC).                                                                                                                                                                                                                                                                                                                                                                                                                          | 10                       |       |
| NET 1.2   | Documentation of zones and network zone interconnections | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Layered Network Defenses                                     | NET-02   | Mechanisms exist to implement security functions as a layered structure that minimizes interactions between layers of the design and avoids any dependence by lower layers on the functionality or correctness of higher layers.                                                                                                                                                                                                                                                                                                                      | 8                        |       |
| NET 1.2   | Documentation of zones and network zone interconnections | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Interconnection Security Agreements (ISAs)                   | NET-05   | Mechanisms exist to authorize connections from systems to other systems using Interconnection Security Agreements (ISAs), or similar methods, that document, for each interconnection:<br>(1) Interface characteristics;<br>(2) Security, compliance and resilience requirements; and;                                                                                                                                                                                                                                                                | 3                        |       |
| NET 1.3   | Network segmentation from safety systems                 | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Boundary Protection                                          | NET-03   | Mechanisms exist to monitor and control communications at the external network boundary and at key internal boundaries within the network.                                                                                                                                                                                                                                                                                                                                                                                                            | 5                        |       |
| NET 1.3   | Network segmentation from safety systems                 | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Isolation of System Components                               | NET-03.7 | Mechanisms exist to employ boundary protections to isolate Technology Assets, Applications and/or Services (TAAS) that support critical missions and/or business functions.                                                                                                                                                                                                                                                                                                                                                                           | 5                        |       |
| NET 1.3   | Network segmentation from safety systems                 | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Separate Subnet for Connecting to Different Security Domains | NET-03.8 | Mechanisms exist to implement separate network addresses (e.g., different subnets) to connect to systems in different security domains.                                                                                                                                                                                                                                                                                                                                                                                                               | 5                        |       |

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Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focus Document: **International Electrotechnical Commission (IEC) 62443 - 2 - 1:2024 - Security for industrial automation and control systems - Part 2 -**

Focus Document URL: <https://webstore.iec.ch/en/publication/62883>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-iec-62443-2-1-2024.pdf>

| FDE #      | FDE Name                                                | Focal Document Element (FDE) Description                                                                                                                                                | STRM Rationale | STRM Relationship | SCF Control                                          | SCF #    | Secure Controls Framework (SCF) Control Description                                                                                                                                                                                                             | Strength of Relationship | Notes |
|------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------|------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------|
| NET 1.3    | Network segmentation from safety systems                | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Network Segmentation (macrosegmentation)             | NET-06   | Mechanisms exist to implement network segmentation within network architectures to isolate Technology Assets, Applications and/or Services (TAAS) from other network resources.                                                                                 | 5                        |       |
| NET 1.4    | Network autonomy                                        | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Fail Safe                                            | SEA-07.3 | Mechanisms exist to implement fail-safe procedures when failure conditions occur.                                                                                                                                                                               | 5                        |       |
| NET 1.5    | Network disconnection from external networks            | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Sensitive / Regulated Data Enclave (Secure Zone)     | NET-06.3 | Mechanisms exist to implement segmentation controls to restrict inbound and outbound connectivity for sensitive and/or regulated data enclaves (secure zones).                                                                                                  | 3                        |       |
| NET 1.6    | Internal network access control                         | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Data Flow Enforcement – Access Control Lists (ACLs)  | NET-04   | Mechanisms exist to implement and govern Access Control Lists (ACLs) to provide data flow enforcement that explicitly restrict network traffic to only what is authorized.                                                                                      | 5                        |       |
| NET 1.6    | Internal network access control                         | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Internal System Connections                          | NET-05.2 | Mechanisms exist to control internal system connections through authorizing internal connections of systems and documenting, for each internal connection, the interface characteristics, security requirements and the nature of the information communicated. | 5                        |       |
| NET 1.7    | Network accessible services                             | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Isolation of System Components                       | NET-03.7 | Mechanisms exist to employ boundary protections to isolate Technology Assets, Applications and/or Services (TAAS) that support critical missions and/or business functions.                                                                                     | 3                        |       |
| NET 1.7    | Network accessible services                             | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Deny Traffic by Default & Allow Traffic by Exception | NET-04.1 | Mechanisms exist to configure firewall and router configurations to deny network traffic by default and allow network traffic by exception (e.g., deny all, permit by exception).                                                                               | 8                        |       |
| NET 1.8    | User messaging                                          | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Data Flow Enforcement – Access Control Lists (ACLs)  | NET-04   | Mechanisms exist to implement and govern Access Control Lists (ACLs) to provide data flow enforcement that explicitly restrict network traffic to only what is authorized.                                                                                      | 8                        |       |
| NET 1.8    | User messaging                                          | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | End-User Messaging Technologies                      | NET-12.2 | Mechanisms exist to prohibit the transmission of unprotected sensitive and/or regulated data by end-user messaging technologies.                                                                                                                                | 5                        |       |
| NET 1.9    | Network time distribution                               | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Clock Synchronization                                | SEA-20   | Mechanisms exist to utilize time-synchronization technology to synchronize all critical system clocks.                                                                                                                                                          | 10                       |       |
| NET 2.1    | Wireless protocols                                      | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Wireless Networking                                  | NET-15   | Mechanisms exist to control authorized wireless usage and monitor for unauthorized wireless access.                                                                                                                                                             | 10                       |       |
| NET 2.1(a) | N/A                                                     | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Secure Baseline Configurations                       | CFG-02   | Mechanisms exist to develop, document and maintain secure baseline configurations for Technology Assets, Applications and/or Services (TAAS) that are consistent with industry-accepted system hardening standards.                                             | 5                        |       |
| NET 2.1(b) | N/A                                                     | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Internal System Connections                          | NET-05.2 | Mechanisms exist to control internal system connections through authorizing internal connections of systems and documenting, for each internal connection, the interface characteristics, security requirements and the nature of the information communicated. | 5                        |       |
| NET 2.1(c) | N/A                                                     | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Secure Baseline Configurations                       | CFG-02   | Mechanisms exist to develop, document and maintain secure baseline configurations for Technology Assets, Applications and/or Services (TAAS) that are consistent with industry-accepted system hardening standards.                                             | 5                        |       |
| NET 2.2    | Wireless network segmentation                           | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Network Segmentation (macrosegmentation)             | NET-06   | Mechanisms exist to implement network segmentation within network architectures to isolate Technology Assets, Applications and/or Services (TAAS) from other network resources.                                                                                 | 10                       |       |
| NET 2.3    | Wireless properties and addresses                       | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Secure Baseline Configurations                       | CFG-02   | Mechanisms exist to develop, document and maintain secure baseline configurations for Technology Assets, Applications and/or Services (TAAS) that are consistent with industry-accepted system hardening standards.                                             | 8                        |       |
| NET 3.1    | Remote access applications                              | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Remote Access                                        | NET-14   | Mechanisms exist to define, control and review organization-approved, secure remote access methods.                                                                                                                                                             | 10                       |       |
| NET 3.2    | Remote access connections                               | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Remote Access                                        | NET-14   | Mechanisms exist to define, control and review organization-approved, secure remote access methods.                                                                                                                                                             | 10                       |       |
| NET 3.2(a) | N/A                                                     | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Remote Access                                        | NET-14   | Mechanisms exist to define, control and review organization-approved, secure remote access methods.                                                                                                                                                             | 3                        |       |
| NET 3.2(b) | N/A                                                     | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Remote Access                                        | NET-14   | Mechanisms exist to define, control and review organization-approved, secure remote access methods.                                                                                                                                                             | 3                        |       |
| NET 3.2(c) | N/A                                                     | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Remote Access                                        | NET-14   | Mechanisms exist to define, control and review organization-approved, secure remote access methods.                                                                                                                                                             | 3                        |       |
| NET 3.2(d) | N/A                                                     | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Remote Access                                        | NET-14   | Mechanisms exist to define, control and review organization-approved, secure remote access methods.                                                                                                                                                             | 3                        |       |
| NET 3.2(e) | N/A                                                     | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Remote Access                                        | NET-14   | Mechanisms exist to define, control and review organization-approved, secure remote access methods.                                                                                                                                                             | 3                        |       |
| NET 3.2(f) | N/A                                                     | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Remote Access                                        | NET-14   | Mechanisms exist to define, control and review organization-approved, secure remote access methods.                                                                                                                                                             | 3                        |       |
| NET 3.2(g) | N/A                                                     | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Remote Access                                        | NET-14   | Mechanisms exist to define, control and review organization-approved, secure remote access methods.                                                                                                                                                             | 3                        |       |
| NET 3.2(h) | N/A                                                     | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Remote Access                                        | NET-14   | Mechanisms exist to define, control and review organization-approved, secure remote access methods.                                                                                                                                                             | 3                        |       |
| NET 3.3    | Remote access termination                               | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Network Connection Termination                       | NET-07   | Mechanisms exist to terminate network connections at the end of a session or after an organization-defined time period of inactivity.                                                                                                                           | 5                        |       |
| NET 3.3    | Remote access termination                               | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Secure Baseline Configurations                       | CFG-02   | Mechanisms exist to develop, document and maintain secure baseline configurations for Technology Assets, Applications and/or Services (TAAS) that are consistent with industry-accepted system hardening standards.                                             | 8                        |       |
| COMP 1.1   | Component hardening                                     | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Secure Baseline Configurations                       | CFG-02   | Mechanisms exist to develop, document and maintain secure baseline configurations for Technology Assets, Applications and/or Services (TAAS) that are consistent with industry-accepted system hardening standards.                                             | 10                       |       |
| COMP 1.2   | Dedicated portable media                                | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Removable Media Security                             | DCH-12   | Mechanisms exist to restrict removable media in accordance with data handling and acceptable usage parameters.                                                                                                                                                  | 8                        |       |
| COMP 1.2   | Dedicated portable media                                | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Portable Storage Devices                             | DCH-13.2 | Mechanisms exist to restrict or prohibit the use of portable storage devices by users on external systems.                                                                                                                                                      | 8                        |       |
| COMP 2.1   | Malware free                                            | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Removable Media Security                             | DCH-12   | Mechanisms exist to restrict removable media in accordance with data handling and acceptable usage parameters.                                                                                                                                                  | 8                        |       |
| COMP 2.1   | Malware free                                            | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Always On Protection                                 | END-04.7 | Mechanisms exist to ensure that anti-malware technologies are continuously running in real-time and cannot be disabled or altered by non-privileged users, unless specifically authorized by management on a case-by-case basis for a limited time period.      | 8                        |       |
| COMP 2.2   | Malware protection                                      | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Malicious Code Protection (Anti-Malware)             | END-04   | Mechanisms exist to utilize anti-malware technologies to detect and eradicate malicious code.                                                                                                                                                                   | 10                       |       |
| COMP 2.3   | Malware protection software validation and installation | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Automatic Antimalware Signature Updates              | END-04.1 | Automated mechanisms exist to update anti-malware technologies, including signature definitions.                                                                                                                                                                | 5                        |       |
| COMP 2.3   | Malware protection software validation and installation | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Malware Protection Mechanism Testing                 | END-04.5 | Mechanisms exist to test anti-malware technologies by introducing a known benign, non-spreading test case into the system and subsequently verifying that both detection of the test case and associated incident reporting occurs.                             | 5                        |       |
| COMP 3.1   | Security patch authenticity/integrity                   | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Software Patch Integrity                             | VPM-05.8 | Mechanisms exist to ensure software and/or firmware patches are: (1) Obtained from trusted sources; and (2) Checked for integrity.                                                                                                                              | 8                        |       |
| COMP 3.2   | Security patch validation and installation              | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Pre-Deployment Patch Testing                         | VPM-05.6 | Mechanisms exist to perform due diligence on software and/or firmware update stability by conducting pre-production testing in a non-production environment.                                                                                                    | 8                        |       |
| COMP 3.3   | Security patch status                                   | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Automated Remediation Status                         | VPM-05.2 | Automated mechanisms exist to determine the state of system components with regard to flaw remediation.                                                                                                                                                         | 8                        |       |
| COMP 3.4   | Security patching retention of security                 | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Test, Validate & Document Changes                    | CHG-02.2 | Mechanisms exist to appropriately test and document proposed changes in a non-production environment before changes are implemented in a production environment.                                                                                                | 8                        |       |
| COMP 3.4   | Security patching retention of security                 | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Pre-Deployment Patch Testing                         | VPM-05.6 | Mechanisms exist to perform due diligence on software and/or firmware update stability by conducting pre-production testing in a non-production environment.                                                                                                    | 5                        |       |
| COMP 3.5   | Security patch mitigation                               | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Compensating Countermeasures                         | RSK-06.2 | Mechanisms exist to identify and implement compensating countermeasures to reduce risk and exposure to threats.                                                                                                                                                 | 5                        |       |
| COMP 3.5   | Security patch mitigation                               | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Deferred Patching Decisions                          | VPM-04.3 | Mechanisms exist to facilitate the deferral of software and/or firmware patches when the disadvantages of applying the patch outweighs the benefits.                                                                                                            | 8                        |       |

**NIST IR 8477-Based Set Theory Relationship Mapping (STRM)**

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: [International Electrotechnical Commission \(IEC\) 62443 - 2 - 1:2024 - Security for industrial automation and control systems - Part 2 -](https://webstore.iec.ch/en/publication/62883)

Focal Document URL: <https://webstore.iec.ch/en/publication/62883>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-iec-62443-2-1-2024.pdf>

| FDE #       | FDE Name                                   | Focal Document Element (FDE) Description                                                                                                                                                | STRM Rationale | STRM Relationship | SCF Control                                                                                            | SCF #    | Secure Controls Framework (SCF) Control Description                                                                                                                                                                                                                                                           | Strength of Relationship | Notes |
|-------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------|--------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------|
| DATA 1.1    | Data classification                        | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Data & Asset Classification                                                                            | DCH-02   | Mechanisms exist to ensure data and assets are categorized in accordance with applicable statutory, regulatory and contractual requirements.                                                                                                                                                                  | 10                       |       |
| DATA 1.2    | Data confidentiality                       | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Data Protection                                                                                        | DCH-01   | Mechanisms exist to facilitate the implementation of data protection controls.                                                                                                                                                                                                                                | 10                       |       |
| DATA 1.2    | Data confidentiality                       | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Sensitive / Regulated Data Protection                                                                  | DCH-01.2 | Mechanisms exist to protect sensitive and/or regulated data wherever it is processed and/or stored.                                                                                                                                                                                                           | 8                        |       |
| DATA 1.2    | Data confidentiality                       | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Data & Asset Classification                                                                            | DCH-02   | Mechanisms exist to ensure data and assets are categorized in accordance with applicable statutory, regulatory and contractual requirements.                                                                                                                                                                  | 8                        |       |
| DATA 1.3    | Safety system configuration mode           | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Safety Assessment                                                                                      | EMB-15   | Mechanisms exist to evaluate the safety aspects of embedded technologies via a fault tree analysis, or similar method, to determine possible consequences of misuse, misconfiguration and/or failure.                                                                                                         | 3                        |       |
| DATA 1.3(a) | N/A                                        | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Configuration Change Control                                                                           | CHG-02   | Mechanisms exist to govern the technical configuration change control processes.                                                                                                                                                                                                                              | 3                        |       |
| DATA 1.3(b) | N/A                                        | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Configuration Change Control                                                                           | CHG-02   | Mechanisms exist to govern the technical configuration change control processes.                                                                                                                                                                                                                              | 3                        |       |
| DATA 1.4    | Data retention policy                      | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Media & Data Retention                                                                                 | DCH-18   | Mechanisms exist to retain media and data in accordance with applicable statutory, regulatory and contractual obligations.                                                                                                                                                                                    | 10                       |       |
| DATA 1.5    | Cryptographic mechanisms                   | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Use of Cryptographic Controls                                                                          | CRY-01   | Mechanisms exist to facilitate the implementation of cryptographic protections controls using known public standards and trusted cryptographic technologies.                                                                                                                                                  | 10                       |       |
| DATA 1.6    | Key management                             | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Cryptographic Key Management                                                                           | CRY-09   | Mechanisms exist to facilitate cryptographic key management controls to protect the confidentiality, integrity and availability of keys.                                                                                                                                                                      | 10                       |       |
| DATA 1.7    | Data integrity                             | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Transmission Integrity                                                                                 | CRY-04   | Cryptographic mechanisms exist to protect the integrity of data being transmitted.                                                                                                                                                                                                                            | 8                        |       |
| DATA 1.7    | Data integrity                             | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | File Integrity Monitoring (FIM)                                                                        | MON-01.7 | Mechanisms exist to utilize a File Integrity Monitor (FIM), or similar change-detection technology, on critical Technology Assets, Applications and/or Services (TAAS) to generate alerts for unauthorized modifications.                                                                                     | 5                        |       |
| USER 1.1    | User identity assignment                   | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Identity & Access Management (IAM)                                                                     | IAC-01   | Mechanisms exist to facilitate the implementation of identification and access management controls.                                                                                                                                                                                                           | 10                       |       |
| USER 1.1    | User identity assignment                   | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Identification & Authentication for Organizational Users                                               | IAC-02   | Mechanisms exist to uniquely identify and centrally Authenticate, Authorize and Audit (AAA) organizational users and processes acting on behalf of organizational users.                                                                                                                                      | 8                        |       |
| USER 1.1    | User identity assignment                   | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Identification & Authentication for Non-Organizational Users                                           | IAC-03   | Mechanisms exist to uniquely identify and centrally Authenticate, Authorize and Audit (AAA) third-party users and processes that provide services to the organization.                                                                                                                                        | 8                        |       |
| USER 1.1    | User identity assignment                   | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Identification & Authentication for Devices                                                            | IAC-04   | Mechanisms exist to uniquely identify and centrally Authenticate, Authorize and Audit (AAA) devices before establishing a connection using bidirectional authentication that is cryptographically- based and replay resistant.                                                                                | 8                        |       |
| USER 1.1    | User identity assignment                   | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Identification & Authentication for Third-Party Technology Assets, Applications and/or Services (TAAS) | IAC-05   | Mechanisms exist to identify and authenticate third-party Technology Assets, Applications and/or Services (TAAS).                                                                                                                                                                                             | 8                        |       |
| USER 1.2    | User identity removal                      | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | User Provisioning & De-Provisioning                                                                    | IAC-07   | Mechanisms exist to utilize a formal user registration and de-registration process that governs the assignment of access rights.                                                                                                                                                                              | 8                        |       |
| USER 1.2    | User identity removal                      | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Account Management                                                                                     | IAC-15   | Mechanisms exist to:<br>(1) Define authorized system account types;<br>(2) Define prohibited system account types; and                                                                                                                                                                                        | 8                        |       |
| USER 1.3    | User identity persistence                  | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Account Management                                                                                     | IAC-15   | Mechanisms exist to:<br>(1) Define authorized system account types;<br>(2) Define prohibited system account types; and                                                                                                                                                                                        | 8                        |       |
| USER 1.3    | User identity persistence                  | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Automated System Account Management (Directory Services)                                               | IAC-15.1 | Automated mechanisms exist to support the management of system accounts (e.g., directory services).                                                                                                                                                                                                           | 5                        |       |
| USER 1.4    | Access rights assignment                   | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Authenticate, Authorize and Audit (AAA)                                                                | IAC-01.2 | Mechanisms exist to strictly govern the use of Authenticate, Authorize and Audit (AAA) solutions, both on-premises and those hosted by an External Service Provider (ESP).                                                                                                                                    | 8                        |       |
| USER 1.4    | Access rights assignment                   | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Automated System Account Management (Directory Services)                                               | IAC-15.1 | Automated mechanisms exist to support the management of system accounts (e.g., directory services).                                                                                                                                                                                                           | 8                        |       |
| USER 1.5    | Least privilege                            | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Role-Based Access Control (RBAC)                                                                       | IAC-08   | Mechanisms exist to enforce Role-Based Access Control (RBAC) for Technology Assets, Applications, Services and/or Data (TAASD) to restrict access to individuals assigned specific roles with legitimate business needs.                                                                                      | 10                       |       |
| USER 1.6    | Software service authentication            | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Authenticate, Authorize and Audit (AAA)                                                                | IAC-01.2 | Mechanisms exist to strictly govern the use of Authenticate, Authorize and Audit (AAA) solutions, both on-premises and those hosted by an External Service Provider (ESP).                                                                                                                                    | 8                        |       |
| USER 1.6    | Software service authentication            | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Identification & Authentication for Third-Party Technology Assets, Applications and/or Services (TAAS) | IAC-05   | Mechanisms exist to identify and authenticate third-party Technology Assets, Applications and/or Services (TAAS).                                                                                                                                                                                             | 8                        |       |
| USER 1.7    | Software services interactive login rights | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Secure Baseline Configurations                                                                         | CFG-02   | Mechanisms exist to develop, document and maintain secure baseline configurations for Technology Assets, Applications and/or Services (TAAS) that are consistent with industry-accepted system hardening standards.                                                                                           | 8                        |       |
| USER 1.7    | Software services interactive login rights | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Authenticate, Authorize and Audit (AAA)                                                                | IAC-01.2 | Mechanisms exist to strictly govern the use of Authenticate, Authorize and Audit (AAA) solutions, both on-premises and those hosted by an External Service Provider (ESP).                                                                                                                                    | 5                        |       |
| USER 1.7    | Software services interactive login rights | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Automated System Account Management (Directory Services)                                               | IAC-15.1 | Automated mechanisms exist to support the management of system accounts (e.g., directory services).                                                                                                                                                                                                           | 5                        |       |
| USER 1.8    | Human user authentication                  | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Secure Baseline Configurations                                                                         | CFG-02   | Mechanisms exist to develop, document and maintain secure baseline configurations for Technology Assets, Applications and/or Services (TAAS) that are consistent with industry-accepted system hardening standards.                                                                                           | 8                        |       |
| USER 1.8    | Human user authentication                  | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Automated System Account Management (Directory Services)                                               | IAC-15.1 | Automated mechanisms exist to support the management of system accounts (e.g., directory services).                                                                                                                                                                                                           | 5                        |       |
| USER 1.8(a) | N/A                                        | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Automated System Account Management (Directory Services)                                               | IAC-15.1 | Automated mechanisms exist to support the management of system accounts (e.g., directory services).                                                                                                                                                                                                           | 5                        |       |
| USER 1.8(b) | N/A                                        | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Automated System Account Management (Directory Services)                                               | IAC-15.1 | Automated mechanisms exist to support the management of system accounts (e.g., directory services).                                                                                                                                                                                                           | 5                        |       |
| USER 1.8(c) | N/A                                        | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Automated System Account Management (Directory Services)                                               | IAC-15.1 | Automated mechanisms exist to support the management of system accounts (e.g., directory services).                                                                                                                                                                                                           | 5                        |       |
| USER 1.9    | Multifactor authentication (MFA)           | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Multi-Factor Authentication (MFA)                                                                      | IAC-06   | Automated mechanisms exist to enforce Multi-Factor Authentication (MFA) for:<br>(1) Remote network access;<br>(2) Third-party Technology Assets, Applications and/or Services (TAAS); and /or<br>(3) Non-console access to critical TAAS that store, transmit and/or process sensitive and/or regulated data. | 10                       |       |
| USER 1.10   | Mutual authentication                      | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Mutual Authentication (MA)                                                                             | IAC-30   | Mechanisms exist to enforce Mutual Authentication (MA) where both sides of a communications channel verify the identity of the other party through certificate exchange.                                                                                                                                      | 10                       |       |
| USER 1.11   | Password protection                        | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Authenticator Management                                                                               | IAC-10   | Mechanisms exist to:<br>(1) Securely manage authenticators for users and devices; and<br>(2) Ensure the strength of authentication is appropriate to the classification of the data being accessed.                                                                                                           | 10                       |       |

| FDE #     | FDE Name                                   | Focal Document Element (FDE) Description                                                                                                                                                | STRM Rationale | STRM Relationship | SCF Control                                                                      | SCF #     | Secure Controls Framework (SCF) Control Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Strength of Relationship | Notes |
|-----------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------|----------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------|
| USER 1.11 | Password protection                        | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Password-Based Authentication                                                    | IAC-10.1  | Mechanisms exist to enforce complexity, length and lifespan considerations to ensure strong criteria for password-based authentication.                                                                                                                                                                                                                                                                                                                                                                             | 8                        |       |
| USER 1.12 | Shared and disclosed/compromised passwords | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Events Requiring Authenticator Change                                            | IAC-10.13 | Mechanisms exist to change authentication credentials:<br>(1) At predefined intervals; and/or<br>(2) Upon suspicion of credential compromise.                                                                                                                                                                                                                                                                                                                                                                       | 8                        |       |
| USER 1.12 | Shared and disclosed/compromised passwords | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Restrictions on Shared Groups / Accounts                                         | IAC-15.5  | Mechanisms exist to authorize the use of shared/group accounts only under certain organization-defined conditions.                                                                                                                                                                                                                                                                                                                                                                                                  | 8                        |       |
| USER 1.13 | User login display information             | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Equal             | Previous Logon Notification                                                      | SEA-19    | Mechanisms exist to configure systems that process, store or transmit sensitive and/or regulated data to notify the user, upon successful logon, of the number of unsuccessful logon attempts since the last successful logon.                                                                                                                                                                                                                                                                                      | 10                       |       |
| USER 1.14 | User login failure displays                | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Authenticator Feedback                                                           | IAC-11    | Mechanisms exist to obscure the feedback of authentication information during the authentication process to protect the information from possible exploitation/use by unauthorized individuals.                                                                                                                                                                                                                                                                                                                     | 5                        |       |
| USER 1.15 | Consecutive login failures                 | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Account Lockout                                                                  | IAC-22    | Mechanisms exist to enforce a limit for consecutive invalid login attempts by a user during an organization-defined time period and automatically locks the account when the maximum number of unsuccessful attempts is exceeded.                                                                                                                                                                                                                                                                                   | 8                        |       |
| USER 1.16 | Session integrity                          | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Equal             | Session Integrity                                                                | NET-09    | Mechanisms exist to protect the authenticity and integrity of communications sessions.                                                                                                                                                                                                                                                                                                                                                                                                                              | 10                       |       |
| USER 1.17 | Concurrent sessions                        | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Equal             | Concurrent Session Control                                                       | IAC-23    | Mechanisms exist to limit the number of concurrent sessions for each system account.                                                                                                                                                                                                                                                                                                                                                                                                                                | 10                       |       |
| USER 1.18 | Screen lock                                | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Equal             | Session Lock                                                                     | IAC-24    | Mechanisms exist to initiate a session lock after an organization-defined time period of inactivity, or upon receiving a request from a user and retain the session lock until the user reestablishes access using established identification and authentication methods.                                                                                                                                                                                                                                           | 10                       |       |
| USER 1.19 | Component authentication                   | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Identification & Authentication for Devices                                      | IAC-04    | Mechanisms exist to uniquely identify and centrally Authenticate, Authorize and Audit (AAA) devices before establishing a connection using bidirectional authentication that is cryptographically- based and replay resistant.                                                                                                                                                                                                                                                                                      | 8                        |       |
| USER 2.1  | Authorization                              | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Access Enforcement                                                               | IAC-20    | Mechanisms exist to enforce Logical Access Control (LAC) permissions that conform to the principle of "least privilege."                                                                                                                                                                                                                                                                                                                                                                                            | 10                       |       |
| USER 2.2  | Separation of duties                       | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Equal             | Separation of Duties (SoD)                                                       | HRS-11    | Mechanisms exist to implement and maintain Separation of Duties (SoD) to prevent potential inappropriate activity without collusion.                                                                                                                                                                                                                                                                                                                                                                                | 10                       |       |
| USER 2.3  | Multiple approvals                         | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Dual Authorization for Change                                                    | CHG-04.3  | Mechanisms exist to enforce a two-person rule for implementing changes to critical Technology Assets, Applications and/or Services (TAAS).                                                                                                                                                                                                                                                                                                                                                                          | 8                        |       |
| USER 2.3  | Multiple approvals                         | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Dual Authorization for Privileged Commands                                       | IAC-20.5  | Automated mechanisms exist to enforce dual authorization for privileged commands.                                                                                                                                                                                                                                                                                                                                                                                                                                   | 8                        |       |
| USER 2.4  | Manual elevation of privileges             | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Dual Authorization for Privileged Commands                                       | IAC-20.5  | Automated mechanisms exist to enforce dual authorization for privileged commands.                                                                                                                                                                                                                                                                                                                                                                                                                                   | 8                        |       |
| EVENT 1.1 | Event detection                            | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Event Log Analysis & Triage                                                      | MON-17    | Mechanisms exist to ensure event log reviews include analysis and triage practices that integrate with the organization's established incident response processes.                                                                                                                                                                                                                                                                                                                                                  | 10                       |       |
| EVENT 1.2 | Event reporting                            | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Security Event Monitoring                                                        | MON-01.8  | Mechanisms exist to review event logs on an ongoing basis and escalate incidents in accordance with established timelines and procedures.                                                                                                                                                                                                                                                                                                                                                                           | 10                       |       |
| EVENT 1.3 | Event reporting interfaces                 | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Contacts With Groups & Associations                                              | GOV-07    | Mechanisms exist to establish contact with selected groups and associations within the security, compliance and resilience communities to:<br>(1) Facilitate ongoing cybersecurity and data protection education and training for organizational personnel;<br>(2) Maintain currency with recommended cybersecurity and data protection practices, techniques and technologies; and<br>(3) Share current cybersecurity and/or data protection-related information including threats, vulnerabilities and incidents. | 10                       |       |
| EVENT 1.4 | Logging                                    | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | System Generated Alerts                                                          | MON-01.4  | Mechanisms exist to generate, monitor, correlate and respond to alerts from physical, cybersecurity, data protection and supply chain activities to achieve integrated situational awareness.                                                                                                                                                                                                                                                                                                                       | 8                        |       |
| EVENT 1.4 | Logging                                    | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Event Log Retention                                                              | MON-10    | Mechanisms exist to retain event logs for a time period consistent with records retention requirements to provide support for after-the-fact investigations of security incidents and to meet statutory, regulatory and contractual retention requirements.                                                                                                                                                                                                                                                         | 8                        |       |
| EVENT 1.5 | Log entries                                | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Content of Event Logs                                                            | MON-03    | Mechanisms exist to configure Technology Assets, Applications and/or Services (TAAS) to produce event logs that contain sufficient information to, at a minimum:<br>(1) Establish what type of event occurred;<br>(2) When (date and time) the event occurred;<br>(3) Where the event occurred;<br>(4) The source of the event;<br>(5) The outcome (success or failure) of the event; and<br>(6) The identity of any user/subject associated with the event.                                                        | 10                       |       |
| EVENT 1.6 | Log access                                 | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Protection of Event Logs                                                         | MON-08    | Mechanisms exist to protect event logs and audit tools from unauthorized access, modification and deletion.                                                                                                                                                                                                                                                                                                                                                                                                         | 10                       |       |
| EVENT 1.7 | Event analysis                             | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Security Event Monitoring                                                        | MON-01.8  | Mechanisms exist to review event logs on an ongoing basis and escalate incidents in accordance with established timelines and procedures.                                                                                                                                                                                                                                                                                                                                                                           | 10                       |       |
| EVENT 1.8 | Incident handling and response             | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Incident Response Operations                                                     | IRO-01    | Mechanisms exist to implement and govern processes and documentation to facilitate an organization-wide response capability for cybersecurity and data protection-related incidents.                                                                                                                                                                                                                                                                                                                                | 10                       |       |
| EVENT 1.8 | Incident handling and response             | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Incident Handling                                                                | IRO-02    | Mechanisms exist to cover:<br>(1) Preparation;<br>(2) Automated event detection or manual incident report intake;<br>(3) Analysis;<br>(4) Containment;<br>(5) Eradication; and<br>(6) Recovery.                                                                                                                                                                                                                                                                                                                     | 8                        |       |
| EVENT 1.9 | Vulnerability handling                     | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Vulnerability Remediation Process                                                | VPM-02    | Mechanisms exist to ensure that vulnerabilities are properly identified, tracked and remediated.                                                                                                                                                                                                                                                                                                                                                                                                                    | 8                        |       |
| AVAIL 1.1 | Continuity management                      | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Business Continuity Management System (BCMS)                                     | BCD-01    | Mechanisms exist to facilitate the implementation of contingency planning controls to help ensure resilient Technology Assets, Applications and/or Services (TAAS) (e.g., Continuity of Operations Plan (COOP) or Business Continuity & Disaster Recovery (BC/DR) playbooks).                                                                                                                                                                                                                                       | 10                       |       |
| AVAIL 1.2 | Resource availability management           | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Capacity & Performance Management                                                | CAP-01    | Mechanisms exist to facilitate the implementation of capacity management controls to ensure optimal system performance to meet expected and anticipated future capacity requirements.                                                                                                                                                                                                                                                                                                                               | 10                       |       |
| AVAIL 1.3 | Failure-state                              | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Restore Within Time Period                                                       | BCD-12.4  | Mechanisms exist to restore Technology Assets, Applications, Services and/or Data (TAASD) within organization-defined restoration time-periods from configuration-controlled and integrity-protected information; representing a known, operational state for the asset.                                                                                                                                                                                                                                            | 8                        |       |
| AVAIL 1.3 | Failure-state                              | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Technology Assets, Applications and/or Services (TAAS) Recovery & Reconstitution | BCD-12    | Mechanisms exist to ensure the secure recovery and reconstitution of Technology Assets, Applications and/or Services (TAAS) to a known state after a disruption, compromise or failure.                                                                                                                                                                                                                                                                                                                             | 8                        |       |
| AVAIL 2.1 | Backup                                     | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Subset Of         | Data Backups                                                                     | BCD-11    | Mechanisms exist to create recurring backups of data, software and/or system images, as well as verify the integrity of these backups, to ensure the availability of the data to satisfy Recovery Time Objectives (RTOs) and Recovery Point Objectives (RPOs).                                                                                                                                                                                                                                                      | 10                       |       |
| AVAIL 2.2 | Backup non-interference                    | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Data Backups                                                                     | BCD-11    | Mechanisms exist to create recurring backups of data, software and/or system images, as well as verify the integrity of these backups, to ensure the availability of the data to satisfy Recovery Time Objectives (RTOs) and Recovery Point Objectives (RPOs).                                                                                                                                                                                                                                                      | 5                        |       |

| FDE #     | FDE Name            | Focal Document Element (FDE) Description                                                                                                                                                | STRM Rationale | STRM Relationship | SCF Control                                                                      | SCF #    | Secure Controls Framework (SCF) Control Description                                                                                                                                                             | Strength of Relationship | Notes |
|-----------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------|----------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------|
| AVAIL 2.3 | Backup verification | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Test Restoration Using Sampling                                                  | BCD-11.5 | Mechanisms exist to utilize sampling of available backups to test recovery capabilities as part of business continuity plan testing.                                                                            | 8                        |       |
| AVAIL 2.4 | Backup media        | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Separate Storage for Critical Information                                        | BCD-11.2 | Mechanisms exist to store backup copies of critical software and other security-related information in a separate facility or in a fire-rated container that is not collocated with the system being backed up. | 5                        |       |
| AVAIL 2.4 | Backup media        | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Cryptographic Protection                                                         | BCD-11.4 | Cryptographic mechanisms exist to prevent the unauthorized disclosure and/or modification of backup information.                                                                                                | 5                        |       |
| AVAIL 2.5 | Backup restoration  | For specific requirement details, buy a licensed copy of the Focal Document at: <a href="https://webstore.iec.ch/en/publication/62883">https://webstore.iec.ch/en/publication/62883</a> | Functional     | Intersects With   | Technology Assets, Applications and/or Services (TAAS) Recovery & Reconstitution | BCD-12   | Mechanisms exist to ensure the secure recovery and reconstitution of Technology Assets, Applications and/or Services (TAAS) to a known state after a disruption, compromise or failure.                         | 8                        |       |